MIL-G-24934

26 SEP 1988

MILITARY SPECIFICATION

GLOVES, FIREMEN'S

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers requirements for a waterproof, heat protective gauntlet glove.

1.2 Classification. The gloves shall be of one type only and in the following sizes (see 6.2):

Schedule of sizes

Extra-Small (XS)
Small (S)
Medium (M)
Large (L)
Extra-Large (XL)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

LAWS AND REGULATIONS

US POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, Attn: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1404).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Applications for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606-5808).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM E 96 - Water Vapor Transmission of Materials
ASTM D-1424 - Tear Resistance of Woven Fabrics by Falling Pendulum (Elmendorf) Apparatus
ASTM D-1785 - Poly(Vinyl Chloride)(PVC) Plastic Pipe

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187).

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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Table I . Physical Requirements

Characteristics	Minimum	Maximum
Wales per inch	20	
Courses per inch	20	
Tear Strength (lbs)		
Wale	10	
Course	8	
Weight (oz/yd2)	8.0 (±0.5)	
Bursting Strength (lbs)	90	
Flame resistance, vertical (wale direction only)		
after flame (sec.)		0.5
char length (inches)		1
after glow (sec.)		15

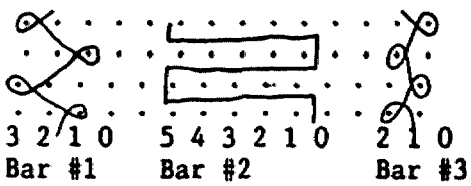
3.3.3.2 Para-aramid. The fiber shall be 1-1/2 denier, 2 inch staple, when tested as specified in 4.4.1.

3.3.3.3 Rayon. The PFR (permanently flame retardant) rayon is manufactured by metering water-insoluble phosphorous-based chemical compounds into viscose solution from which rayon fibers are spun. The fibers contain the flame retardant which is highly dispersed in a matrix of PFR cellulose. The fiber shall be a nominal 1.5 - 2.0 denier and nominal 1.5 - 2.0 inch staple cut length, when tested as specified in 4.4.1.

3.3.3.4 PBI. The fiber shall be 1-1/2 denier, 2 inch staple, when tested as specified in 4.4.1

3.3.3.5 Color The color shall be natural.

3.3.3.6 Knit construction. The guide bar stitch pattern shall be as follows when tested as specified in 4.4.1:



Front guide bar #1: 1-0/2-3
Middle guide bar #2: 0-0/5-5
Back guide bar #3: 1-2/1-0

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3.4 Design. The five fingered leather gauntlet glove shall be of a Gunn cut design with straight thumb. A microporous film glove insert between the inseamed leather shell and napped knitted para-aramid/PFR rayon/PBI sack liner provides water resistance. The back of the glove shall have an elastic webbing takeup and the attached gauntlet shall be unlined.

3.4.1 Figures. Figures are furnished for information purposes only. When inconsistencies exist between the written specification and the figure, the written document shall govern.

3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government (see 6.3). The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way, except that additional notches for use during construction are permitted on the working patterns. Unless otherwise specified in Table II, the shell patterns provide a 3/16 inch seam allowance and the liner patterns 1/4 inch.

3.5.1 Pattern parts. The component parts shall be cut from the materials as specified, and in accordance with the following list of pattern parts:

Material	Pattern parts	Cut parts
Split leather	Back, shell	2
	Palm, shell	2
	Thumb, shell	2
	Fingers, shell	2
	Gauntlet, shell	2
Knit: para-aramid/PFR rayon/PBI	Back, liner	2
	Palm, liner	2
	Thumb, liner	2
	Fingers, liner	2

3.6 Construction.

3.6.1 Stitches, seams, and stitching. Stitch, seam, and stitching types specified in Table II shall conform to FED-STD-751. Whenever two or more methods for seam or stitch types are given for the same part of the operation, any one may be used. The use of a trimmer attachment on the seving machine for trimming leather or for trimming the lining is prohibited. Seam allowances shall be maintained with seams properly sewn so that no raw edges, runoffs, twists, pleats or open seams will result. In order to minimize the size of the needle hole in the leather, the needle used in the seaming operations shall be of the smallest size which will accommodate the specified size of thread.

3.6.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table II.

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TABLE II - CONSTRUCTION OF GLOVES

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
1.	Cutting. a. Before cutting the leather, all skins shall be examined for defects which affect the appearance of the end item. The skins shall be dampened with tap water only, and the lengthwise stretch (from butt end to head end) shall be pulled out before cutting. The skins shall be stretched in the width direction, however, sufficient stretch shall remain to prevent constriction or binding across the hand and knuckles in the finished glove when fingers are doubled under to make a fist. b. The length of the glove pattern shall run parallel with the length of the hide. c. Cut liner in accordance with patterns which show directional lines for cutting and notches for proper assembly. The napped side shall be the finished side of liner.				
2.	Replacement of defective components. During the cutting and manufacturing process, components having material defects or damages that are classified as defects in section 4 shall be removed from production and replaced with non-defective and properly matched components.				
3.	Marking. Mark, ticket, or bundle all component leather cut parts to insure a uniform shade and size for proper assembly throughout the gloves (see 3.7.1).				
4.	Prepare outershell. a. Following the marks on pattern, center and stitch a 3 1/2 to 4 inch piece of elastic webbing held under tension to the inside of back shell. The elastic shall be stretched to its full length so that the shirrings on the outside of back, where the webbing is attached, shall be evenly spaced and the full amount of the stretch in the webbing is utilized.	304	SSau-1	6-8	

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TABLE II - CONSTRUCTION OF GLOVES

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
	b. Stitch fingertip tab extensions of glove insert along curved seam allowance of shell fingers. To maintain the waterproof integrity of the glove no stitches shall penetrate the seal or area inside the seal of the glove insert.	301	Smlr to SSa-1	8-10	
	c. Turn glove into the finished position working out fingertips. Place tip of turning device on fingertip of shell fabric using caution to avoid contact with insert when turning.				
7.	Attach gauntlet and hem glove.				
	a. Hem top edge of gauntlet, turning under 3/8 inch while catching the combination label on inside of back of glove gauntlet. Stitching shall be set 3/16 to 5/16 inch from folded edge. The label shall be positioned so that the warning inscription is facing the wearer. The stitching shall not be through the printing.	301	EFa-1	8-10	
	b. With edges aligned, stitch side seam of gauntlet.	301	SSa-1	8-10	
	c. With the finished sides of the leather facing and the gauntlet seam matching the 4th finger seam on the side of the glove, stitch gauntlet to the glove ensuring that the liner and glove insert are caught in the seaming.	301	SSa-1	8-10	
	d. As an alternate, operation 7a may be performed after operation 7 b and c.	301	EFa-1	8-10	
8.	Forming and pairing.				
	a. The gloves shall be dampened, layed off and formed on steam or electrically heated boards to provide proper shape and appearance. The board shall be properly sized to match the glove size.				

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3.8 Measurements. The finished glove shall conform to the dimensions specified in Table III. All measurements and tolerances are expressed in inches. All measurements shall be taken without distorting the finished glove.

Table III - Finished measurements

Size	Overall Length (A)	Thumb Length (B)	Width (C)	Palm Length (D)	Gauntlet Opening (E)
XS	12-3/8	2-7/8	4-1/4	9	5-1/2
S	12-3/4	3	4-1/2	9-3/8	5-1/2
M	13-1/8	3-1/8	4-3/4	9-3/8	6
L	13-1/2	3-1/4	5	9-3/4	6
XL	13-7/8	3-3/8	5-1/4	9-3/4	6-1/2
Tolerance	(± 1/2)	(± 1/4)	(± 1/4)	(±1/4)	(± 1/4)

NOTE: Refer to Figure 1 for A, B, C, D and E Measurements.

A. Overall length. Measurement is taken from the tip of middle finger to the top edge of the gauntlet opening.

B. Thumb length. Measurement is taken from the crotch to the tip of thumb.

C. Width. Measurement is taken from folded edge to folded edge on the palm side, at the crotch of the thumb.

D. Palm length. Measurement is taken from the bottom edge of the gauntlet cuff to seam tip of second finger.

E. Gauntlet opening. Measurement is taken across top edge of the gauntlet from folded edge to folded edge.

NOTE: Length dimensions in single pair of gloves shall not vary more than 1/4 inch.

3.9 Workmanship. After completion of the final assembly, the gloves shall be thoroughly cleaned and all thread scraps, lint, and foreign matter shall be removed. The gloves shall not contain any spot, stain, cut, abraded area, tear, mend, needle chew, or ragged edge. The gloves shall conform the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Table IV - Component tests

Component	Characteristic	Requirement paragraph	Test method
Film insert	Water permeability		
	After exposure to aircraft fluid	3.3.4	5516 2/ 3/
	Weight	3.3.4	1/
	Moisture vapor transmission rate		
	Procedure B	3.3.4	ASTM E96 5/
	Procedure BW	3.3.4	ASTM E96 6/
Knit liner	Yarn blend	3.3.3.1	1/
	Weight	3.3.3.1	5041
	Yarn count	3.3.3.1	4052
	Wales/Courses	3.3.3.1	5070
	Bursting strength	3.3.3.1	5120 4/
	Tear strength	3.3.3.1	ASTM D1424
	Flame resistance	3.3.3.1	5903
	Fiber		
	Para-aramid		
	Identification, staple, denier	3.3.3.2	1/
	PFR rayon		
	Identification, staple, denier	3.3.3.3	1/
	PBI		
	Identification, staple, denier	3.3.3.4	1/
	Knit	3.3.3.6	Visual
Leather shell	Thickness	3.3.1	1011 of FED-STD-311
	Shrinkage temperature	3.3.1	7011 of FED-STD-311
	Silicone identification	3.3.1.1	1/
	Water absorption	3.3.1.1	8111.1 of FED-STD-311 7/

1/ Unless otherwise specified, a certificate of compliance is required and will be accepted for the stated requirement.

2/ The water permeability shall be measured as specified in Method 5516 of FED-STD-191. The outer surface of the insert shall face the water. An unfinished nylon tafetta restraint cloth conforming to type III, class 1 of MIL-C-21852 shall be placed across the opposite surface of the insert. No area of the restraint cloth shall be used more than once. The hydrostatic head shall be 50 cm and shall be held for 3 minutes. The report shall only include measurement of the appearance of water drops. Leakage is defined as the appearance of water any place within the 4-1/2 inch diameter test area. The test may be performed using any test device which tests the same specimen area at the equivalent pressure using an appropriate size specimen. In cases of dispute, the apparatus described in Method 5516 of FED-STD-191 shall be used.

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4.4.2.1 In-process examination of liner/insert assembly. The glove liner and PTFE insert assembly shall be examined after the completion of operation 6b of Table I. The sample unit shall be 1 glove and the selection shall be by pairs. The lot size shall be expressed in the same terms as the sample. The inspection and acceptable quality levels (AQL's) expressed as defects per 100 units for visual examinations shall be as follows:

	AQL	Inspection level
Major	2.5	II
Total (major and minor combined)	6.5	II

NOTE: If one or more critical defects are found during sampling inspection, the lot shall be rejected. For each lot found to be acceptable by the sampling plan, the balance of the lot shall be inspected for critical defects only. Any critical defect found shall be removed from the lot and replaced with a conforming item.

The gloves shall be examined for the defects listed below:

Defect	Classification		
	Critical	Major	Minor
PTFE insert:			
a. Missing	X		
b. Any hole, cut, tear, mend or burn	X		
Adhesive at fingertips:			
a. Missing		X	
b. Not located as specified			X
c. Incompatible, i.e., boardy, tacky, uneven, lumpy or partial delamination etc....			X
Stitching insert fingertip tabs to glove shell:			
a. Omitted			X
b. Penetrating seal or area inside seal	X		

4.4.3 Examination of the end item. Examination of the end item shall be in accordance with 4.4.4 through 4.4.5.

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Table V. End item visual defects

Examine	Defect	Classification Major	Minor
IV. Components and assembly	a. Any component not as specified or any defect of assembly	X	
	b. Any component, component part, or required operation omitted, or any operation improperly performed	X	
	c. Face (napped side) of the cloth not on the exposed surface of glove interior		X
	d. Insufficient stretch across the back of hand when fist is clenched causing constriction	X	
	e. Fingers distorted or twisted:		
	1. Seriously affecting serviceability	X	
	2. Affecting serviceability, but not seriously comfort		X
V. Seams and stitching	a. Accuracy of seaming:		
	1. Any seam, hem, or attachment of any part puckered, pleated, or caught in an unrelated operation or stitching	X	
	2. Any seam allowance not as specified	X	
	b. Gage of stitching: 2/		
	1. Any stitching irregular, unevenly gaged or not gaged as specified, or not uniform	X	
	c. Any open seam 3/	X	
	d. Seam and stitch type:		
	1. Not specified seam or stitch		X
	e. Stitch tension:		
	1. Any loose stitching 4/	X	
	2. Any tight stitching resulting in breaking of the stitches when normal pull is applied 5/	X	
	f. Stitches per inch: 6/		
	1. Stitches more or less than specified:		
	- by more than two stitches	X	
	- by one or two stitches		X
VI. Pairing	a. Not right and left glove of the same size, or wide variation in the appearance	X	
	b. Not tacked or joined together as specified		X
VII. Label	a. Missing, incomplete, or illegible, or size incorrect	X	
	b. Improperly located		X

1/ One or two single pick runs on the innerside of a glove shall not be classified.

2/ The gage of stitching defect shall be classified only when the condition exists on the major portion of the seam.

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4.4.7 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 in accordance with MIL-STD-105.

Examine	Defect
Finished dimensions	Length, width,or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation-packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Each pair of gloves shall be inserted within a snug fitting clear polyethylene bag of 0.00125 inch (+ 25 percent) gage film. One end of the bag shall not be sealed. One end and the two sides shall be sealed. The heat-sealed seams shall be straight, continuous, and parallel to each other and to the edges of the polyethylene bag. The heat sealed seams, as applicable, shall be as close as possible to the edge of the bag. The bag shall be fabricated from polyethylene film tubing or sheeting. No glove shall be folded, deformed, distorted, or bent in the bagged condition, nor shall the polyethylene bag be damaged.

5.1.2 Level C. Gloves shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving authority. The package and quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.2).

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The pallet shall be 4-way, type IV; type V, class 1, size 2; or type VIII, fabricated from wood group I, II, III or IV, grade A of NN-P-71, or 4-way, style 1, size A, type I, class 1 fabricated from wood groups specified of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

6. NOTES

6.1 Intended use. The gloves covered by this specification are intended to be worn by firefighters and repair personnel combating structural fires and performing emergency repairs.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2)
c. When first article sample is required, (see 3.2) the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quantity, and testing and approval of the first article.
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
e. When palletization is required (see 5.4).

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids.

6.4 Glove insert. A glove insert conforming to the requirements of 3.3.4 may be obtained from W.L. Gore and Assoc., Inc., Elkton, MD (Style No. F1 Adult).

6.5 Subject term (key word) listing.
Firefighter glove
Gunn cut gauntlet glove
Leather
Liner, knitted, para-aramid
Polytetraflouroethylene film

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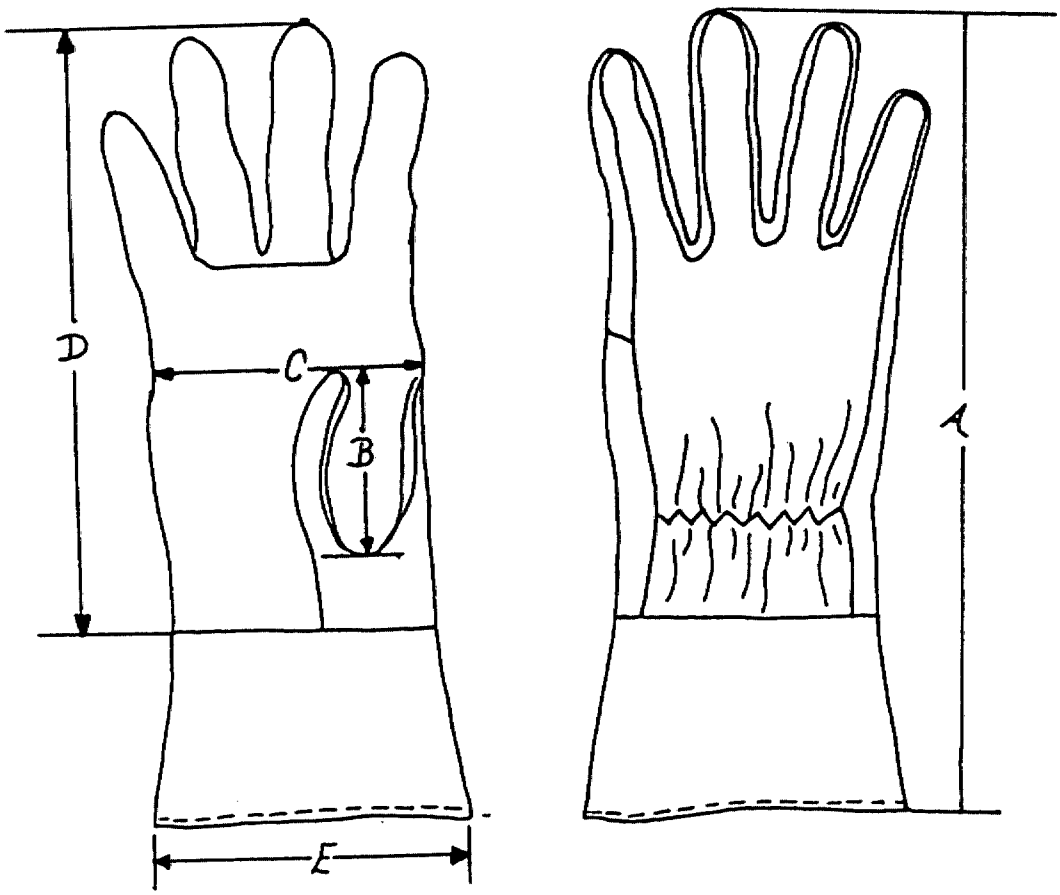


FIGURE 1

GLOVES, FIREMEN'S

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